Chapter 5: Electrostatics Fields in Material

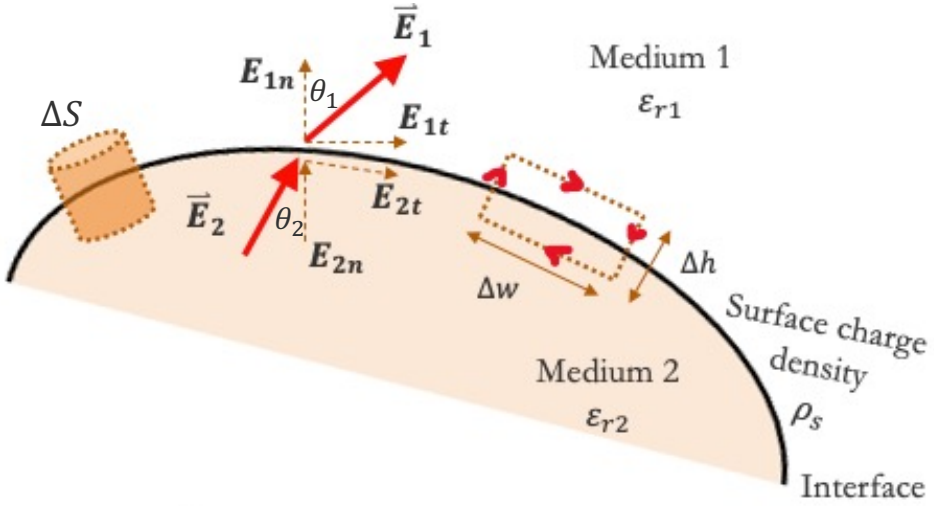
The following subjects will be discussed

- Electrostatic Boundary Conditions.

Electrostatic Boundary Conditions

The relation between the electric field vector at the interface between two different media is given by the "boundary conditions".

To relate the tangential components across the interface, evaluate the closed line integral around some contour.



$$\oint \vec{E} \cdot \overrightarrow{dl} = 0 \;\rightarrow\; E_{1t}\Delta w + E_{1n}\frac{\Delta h}{2} + E_{2n}\frac{\Delta h}{2} - E_{2t}\Delta w - E_{1n}\frac{\Delta h}{2} - E_{2n}\frac{\Delta h}{2} = 0$$

The interface is defined as $\lim_{\Delta h \to 0}$ ➡ $E_{1t} = E_{2t}$ and $\frac{D_{1t}}{\varepsilon_{r1}} = \frac{D_{2t}}{\varepsilon_{r2}}$

Tangential component of E continuous

To relate the normal components across the interface, evaluate Gauss law over some enclosure.

$$\oint_S \vec{D} \cdot \overrightarrow{dS} = Q_{enclosed} \;\rightarrow\; D_{1n}\Delta S - D_{2n}\Delta S = \rho_s \Delta S \;\Rightarrow\; D_{1n} = D_{2n} + \rho_s$$

For no surface charge $D_{1n} = D_{2n}$ and $\varepsilon_{r1}E_{1n} = \varepsilon_{r2}E_{2n}$ Also $\frac{\tan\theta_1}{\tan\theta_2} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}$

Normal component of D continuous

Examples

(1) The x-y plane (at $z = 0$) separates two dielectric materials. The dielectric constant of region 1 ($z > 0$) is 4 and that of region 2 ($z < 0$) is 3. Given,

$$\vec{E}_1 = 5\,\vec{a}_x - 2\,\vec{a}_y + 3\,\vec{a}_z \quad \text{(V/m)}$$

find the electric field in region 2. $\vec{E}_{1t} = 5\,\vec{a}_x - 2\,\vec{a}_y$ $\vec{E}_{1n} = 3\,\vec{a}_z$

$\vec{E}_{2t} = \vec{E}_{1t} = 5\,\vec{a}_x - 2\,\vec{a}_y$ $\vec{E}_{2n} = \frac{\varepsilon_{r1}}{\varepsilon_{r2}}\vec{E}_{1n} = \frac{4}{3}3\,\vec{a}_z = 4\,\vec{a}_z$ $\vec{E}_2 = 5\,\vec{a}_x - 2\,\vec{a}_y + 4\,\vec{a}_z$ (V/m)

Find angles with the normal.

$$\theta_1 = \tan^{-1}\frac{|E_{1t}|}{|E_{1n}|} = \tan^{-1}\frac{\sqrt{29}}{3} = 60.9°$$

$$\theta_2 = \tan^{-1}\left(\frac{\varepsilon_{r2}}{\varepsilon_{r1}}\tan\theta_1\right) = 53.4\,°$$

(2) The region $\rho \leq 3$ is filled with a dielectric that has a dielectric constant of 2.5. The rest of space is air. The electric field in the dielectric is $\vec{E} = 2\,\vec{a}_\rho + 5\,\vec{a}_\varphi - 4\,\vec{a}_z$ (V/m), find the flux density in air.

$$\vec{D} = 44.25\,\vec{a}_\rho + 44.25\,\vec{a}_\varphi - 35.4\,\vec{a}_z \quad \text{(nC/m}^2\text{)}$$

Examples

(3) A parallel-plate capacitor, initially charged as shown. Then a slab of dielectric is inserted.

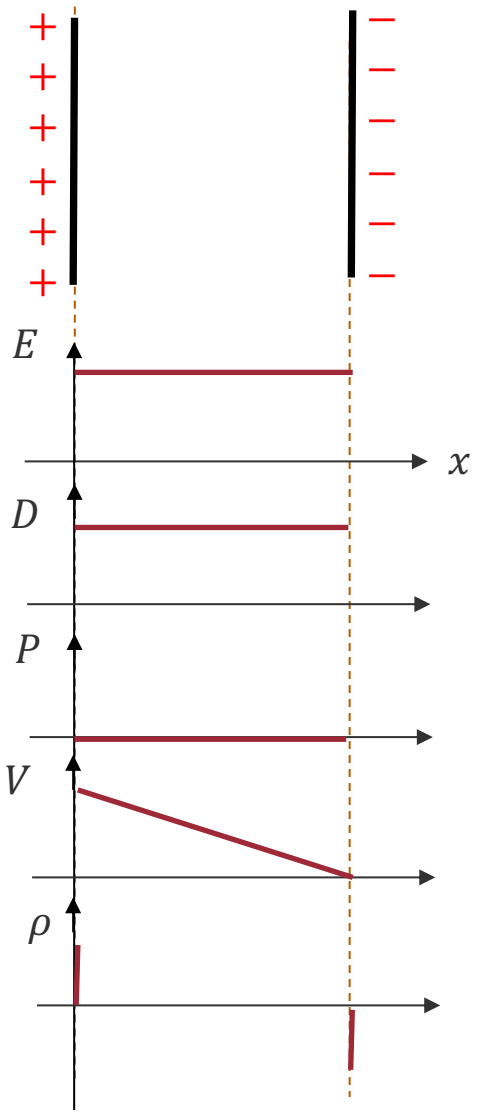
Electric Field
V/m

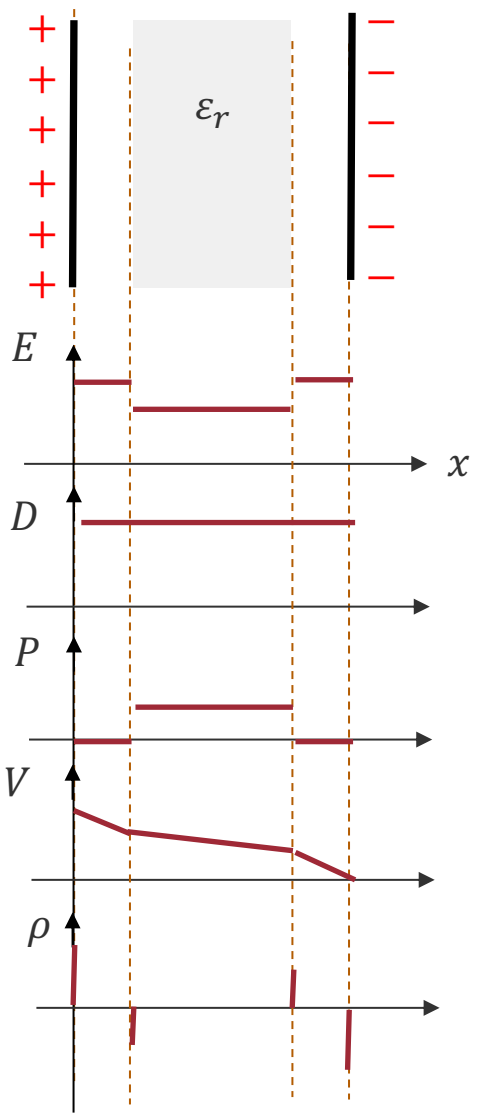
Flux Density
C/m^2

Polarization
C/m^2

Potential
V

Charge density
C/m^3





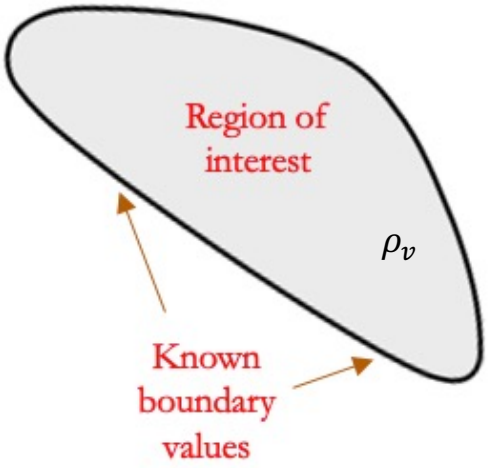
Chapter 6: Electrostatics Boundary-value Problems

The following subjects will be discussed

- Poisson's and Laplace's Equations for Potential.
- Analytical Solution Method.
- Resistance and Capacitance Calculations.

Poisson's and Laplace's Equations

Given the electric potential distribution over the boundaries of a certain region of interest (boundary conditions), it is possible to solve for the electric potential distribution inside that region.



Starting with the basic field relations,

$$\vec{\nabla}\cdot\vec{D}=\rho_v \;\rightarrow\; \vec{\nabla}\cdot\varepsilon\,\vec{E}=\rho_v \;\rightarrow\; \vec{\nabla}\cdot\varepsilon\,(-\vec{\nabla}V)=\rho_v$$

Assuming a homogeneous material, the following equation is obtained for the electric potential. $\vec{\nabla}\cdot(\vec{\nabla}V)=-\rho_v/\varepsilon$

which can be written as, $\vec{\nabla}^2 V=-\frac{\rho_v}{\varepsilon}$ Poisson's Equation $\vec{\nabla}^2 V=0$ Laplace's Equation In case no free charges are present inside region

Rectangular coordinates $\frac{\partial^2 V}{\partial x^2}+\frac{\partial^2 V}{\partial y^2}+\frac{\partial^2 V}{\partial z^2}=-\frac{\rho_v}{\varepsilon}$ $\frac{\partial^2 V}{\partial x^2}+\frac{\partial^2 V}{\partial y^2}+\frac{\partial^2 V}{\partial z^2}=0$

Cylindrical coordinates $\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial V}{\partial\rho}\right)+\frac{1}{\rho^2}\frac{\partial^2 V}{\partial\varphi^2}+\frac{\partial^2 V}{\partial z^2}=-\frac{\rho_v}{\varepsilon}$ $\frac{1}{\rho}\frac{\partial}{\partial\rho}\left(\rho\frac{\partial V}{\partial\rho}\right)+\frac{1}{\rho^2}\frac{\partial^2 V}{\partial\varphi^2}+\frac{\partial^2 V}{\partial z^2}=0$

Analytical Solution Method

Before starting the solution, these parameters need to be identified. (1) Coordinate system (based on structure) (2) Number of dimensions (based on potential variation) (3) Boundary conditions and (4) Poisson or Laplace (based on charge density).

Procedure

(1) Find a general solution of the second order differential equation (Poisson or Laplace). Use methods of integration.
(2) Apply boundary conditions to find a particular solution for the potential function.
(3) Once the V function is obtained other quantities such as $\vec{E}$, $\vec{D}$, C, R can be obtained. Refer to the equation below.

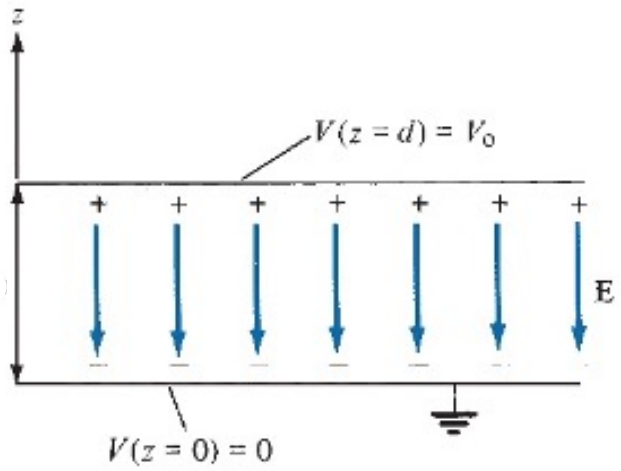
Electric Field $\vec{E} = -\vec{\nabla}V$ Flux Density $\vec{D} = \varepsilon\,\vec{E} = -\varepsilon\,\vec{\nabla}V$

Resistance $R = \dfrac{\int_L \vec{E}\cdot\vec{dl}}{\sigma\int_S \vec{E}\cdot\vec{dS}}$ Capacitance $C = \dfrac{\varepsilon\oint \vec{E}\cdot\vec{dS}}{\int_L \vec{E}\cdot\vec{dl}}$

Note that $RC = \dfrac{\varepsilon}{\sigma}$

Examples

(1) Find potential distribution.



The problem is one-dimensional in rectangular coordinates. Laplace's equation becomes,

$$\frac{\partial^2 V}{\partial z^2} = 0 \quad \rightarrow V(z) = C_1 z + C_2$$

Boundary conditions: $V(0) = 0; \quad V(d) = V_o$

The solution becomes, $$V(z) = \frac{V_o}{d} z$$

$$\vec{E} = -\vec{\nabla} V = -\frac{V_o}{d}\,\vec{a}_z$$

$$\vec{D} = -\frac{\varepsilon\, V_o}{d}\,\vec{a}_z$$

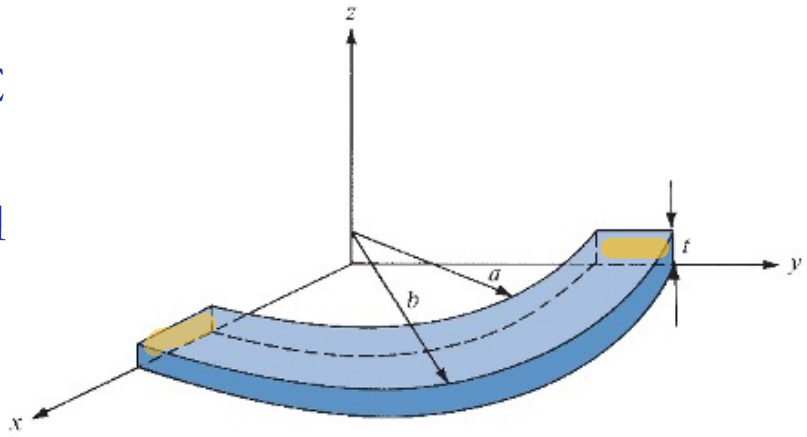
$$\rho_s = \vec{D} \cdot \vec{a}_n = -\frac{\varepsilon\, V_o}{d} \quad \rightarrow \quad Q = \frac{\varepsilon\, A}{d} V_o$$

Examples

(2) Find the resistance and the capacitance of the given structure between the two indicated surface.

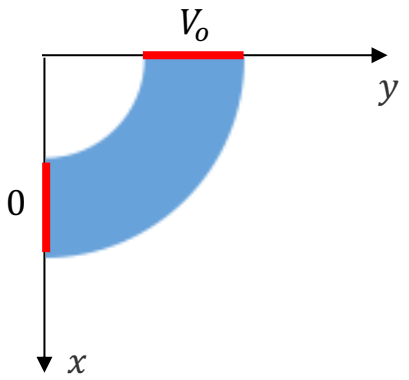
The top view gives a clearer picture. Assign BC to each surface as shown.

The problem is one-dimensional in cylindrical coordinates. Laplace's equation becomes,



$$\frac{1}{\rho^2}\frac{\partial^2 V}{\partial \varphi^2} = 0 \rightarrow \frac{\partial^2 V}{\partial \varphi^2} = 0 \rightarrow V(\varphi) = C_1\varphi + C_2$$

Boundary conditions: $V(0) = 0; \quad V\left(\frac{\pi}{2}\right) = V_o$



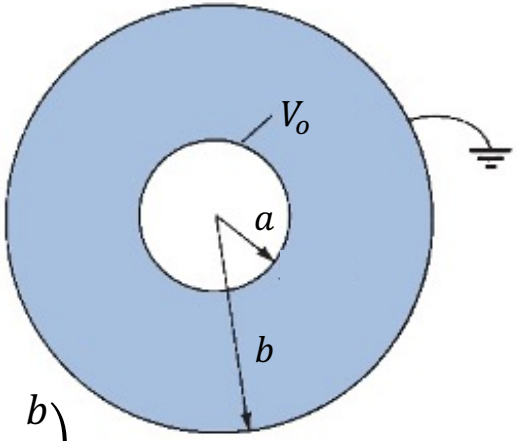
The solution becomes, $V(\varphi) = \frac{2V_o}{\pi}\varphi \qquad \vec{E} = -\vec{\nabla}V = -\frac{2V_o}{\rho\pi}\vec{a}_\varphi$

$$I = \sigma \int_S \vec{E}\cdot\overrightarrow{dS} = \frac{2\sigma V_o}{\pi}\int_0^t\int_a^b \frac{d\rho dz}{\rho} = \frac{2\sigma V_o t}{\pi}\ln\left(\frac{b}{a}\right)$$

$$R = \frac{\int_L \vec{E}\cdot\overrightarrow{dl}}{\sigma\int_S \vec{E}\cdot\overrightarrow{dS}} = \frac{\pi}{2\sigma t\ln\left(\frac{b}{a}\right)} \qquad C = \frac{\varepsilon}{R\sigma} = \frac{2\varepsilon t\ln\left(\frac{b}{a}\right)}{\pi}$$

Examples

(3) Find the resistance and the capacitance of the given structure between the inside and outside surfaces.



The problem is one-dimensional in spherical coordinates. Laplace's equation becomes,

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right)=0 \rightarrow \frac{\partial}{\partial r}\left(r^2\frac{\partial V}{\partial r}\right)=0 \quad \rightarrow \quad V(r)=\frac{-C_1}{r}+C_2$$

Boundary conditions: $V(a)=V_o;\ \ V(b)=0$

The solution becomes, $V(r)=\dfrac{-ab\,V_o}{(a-b)}\dfrac{1}{r}+\dfrac{a\,V_o}{(a-b)}=\dfrac{a\,V_o}{(a-b)}\left(1-\dfrac{b}{r}\right)$

$$\vec{E}=-\vec{\nabla}V=-\frac{ab\,V_o}{(a-b)}\frac{1}{r^2}\ \vec{a}_r$$

$$Q=\varepsilon\int_S \vec{E}\cdot\overrightarrow{dS}=-\frac{\varepsilon\,ab\,V_o}{(a-b)}\int_0^{2\pi}\int_a^{\pi}\frac{1}{r^2}r^2\sin\theta\,d\theta\,d\varphi=\frac{4\pi\varepsilon\,ab\,V_o}{(b-a)}$$

$$C=\frac{\varepsilon\oint\vec{E}\cdot\overrightarrow{dS}}{\int_L\vec{E}\cdot\overrightarrow{dl}}=\frac{Q}{V}=\frac{4\pi\varepsilon\,ab}{(b-a)} \qquad R=\frac{\varepsilon}{C\sigma}=\frac{(b-a)}{4\pi\sigma ab}$$

Q & A

Send me an email if you have any questions.